Verdant Coil — Phase 1.3: Tile System & Modules (Design Plan)

Version lock: Godot 4.4.1
Documents aligned: Verdant Coil Plan (Phase 1.3 scope), Scene Architecture, UI Design, Game Elements.

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1) Goals & Non-Goals

Goals (end of 1.3)

- Playable coil using **TileMap-only** tiles with metadata, no instanced tile scenes.
- Implement and verify first 5 modules:
- **Flesh Tile** (walkable base)
- **Acid Pool** (damage per entry/turn; mitigated by Hardened Skin)
- **Sticky Web** (adds movement delay; mitigated by future mobility upgrade)
- **Digest Wall** (blocks unless Acid Sac is active; later: consumes Acid Reserve)
- **Heartroot** (goal; triggers win condition)
- Define a **single, deterministic interaction order** (per turn) and use it in Explore Mode.
- Establish a **stable metadata schema** that Builder Mode will paint/read unchanged.
- Provide **QA test maps** and a checklist that Builder Mode will reuse.

Non-Goals (defer)

- Instanced tile scenes for hazards (reserved for special/animated entities post-MVP).
- Builder UI, biomass costs, undo/redo (Phase 1.4–2.1).
- Economy hooks (Nutrient/Sporeprint), save/load (1.5).

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2) Architectural Decisions

A. Tile system approach (single source of truth)

- **Use TileMap layers + TileSet metadata only** for Phase 1.3 hazards/terrain.
- Crawler queries TileMap → TileData.custom_data to decide pass/block/effects.
- Reserve instanced scenes for **non-tile entities** later (e.g., turrets with timing, multi-tile props).

B. Coordinate & origin conventions

- **Grid:** `Grid.to_world(tile, align_to_center = true)` for agents (Crawler, pickups) → agents sit in the **center** of a cell.
- **TileMap visuals:** naturally drawn **top-left anchored to cell** by Godot; no change needed.
- **Implication:** We are no longer placing hazard scenes, so the prior “origin in the middle” problem goes away. If we later place a non-TileMap scene into a cell (e.g., Heartroot VFX), we’ll center it with `align_to_center = true`.

C. Layers (indices fixed for 1.3)

- **Layer 0 — Base:** Flesh, Digest Wall, Acid Pool, Sticky Web (visual + metadata).
- **Layer 1 — Overlay:** `is_spawn`, `is_goal` (Heartroot marker), optional indicators.
- **Layer 2 — FX (optional):** reserved for polish; no logic in 1.3.

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3) Metadata Schema (TileSet → TileData.custom_data)

Shared keys (present where relevant): - `tile_type: String` — one of: `"flesh" | "acid" | "sticky" | "digest_wall" | "heartroot"`. - `walkable: bool` — can the crawler enter? (e.g., Flesh=true, Digest Wall=false unless upgrade) - `hazard: String` — `"acid" | "sticky" | "none"` (for quick routing without stringly-typed code paths). - `damage_per_step: int` — acid damage applied on entry/turn (default 0). - `slow_ticks: int` — number of turns of slowdown or additional delay (default 0). - `digestible: bool` — if true, Acid Sac permits entry; later: consumes Acid Reserve. - `is_goal: bool` — overlay marker (true only on Heartroot overlay tile). - `is_spawn: bool` — overlay marker for crawler spawn.

Notes - Values live entirely in the TileSet and are paintable by Builder Mode. - Crawler/ExploreMode simply **reads** and applies.

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4) Turn & Interaction Order (deterministic)

For each attempted move (one player input): 1) **Pre-check (blocking):** bounds → base tile present → `walkable` check → upgrade gating (`digestible` + Acid Sac). If any fails, movement is blocked. 2) **Movement:** tween to destination. 3) **On-enter effects (tile at destination):** - Acid damage (`damage_per_step`), mitigations first (Hardened Skin), then apply damage. - Sticky slow (`slow_ticks`) — register a movement cooldown (Phase 1.3 can fake with tween time or a per-move lock). 4) **Goal check:** if `is_goal` (via overlay) is true at the new tile → trigger win.

This order is documented and will be reused by Builder Mode's validator in Phase 1.4–2.2.

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5) Staged Work Plan & Checkpoints (No code here)

Stage 1 — Foundation & TileSet setup

Deliverables - Confirm TileMap layers (0: Base, 1: Overlay) and lock their indices. - Update TileSet to include custom_data keys above for each tile type. - Create a thin test map: clear patches of each tile + a single Heartroot overlay tile + one spawn. **Verification** - In editor: inspect TileData custom_data; ensure keys exist and defaults are sane.

Stage 2 — Flesh Tile (walkable baseline)

Deliverables - Flesh tiles painted with `tile_type:"flesh"`, `walkable:true`, others default/none. **Verification** - Crawler can traverse a corridor of Flesh; movement tween length unchanged.

Stage 3 — Acid Pool (damage)

Deliverables - `tile_type:"acid"`, `walkable:true`, `hazard:"acid"`, `damage_per_step:n`. - Hardened Skin mitigation rule captured in Explore Mode logic (read-only upgrade check). **Verification** - Test map with a 3-tile acid run; confirm damage with/without Hardened Skin toggled.

Stage 4 — Sticky Web (slow)

Deliverables - `tile_type:"sticky"`, `walkable:true`, `hazard:"sticky"`, `slow_ticks:n`. - Temporary slow effect implementation (e.g., increase tween duration or add a per-move cooldown n turns). **Verification** - Entering Sticky adds noticeable delay vs Flesh; confirm delay removed when future mobility upgrade is active (stub: toggled flag with no UI elaboration yet).

Stage 5 — Digest Wall (upgrade-gated block)

Deliverables - `tile_type:"digest_wall"`, `walkable:false`, `digestible:true`. - Enter allowed **only** when Acid Sac is active; otherwise blocked. - (Future 1.5): consume Acid Reserve on pass. **Verification** - Corridor blocked by Digest Wall → cannot pass; toggle Acid Sac → can pass.

Stage 6 — Heartroot (goal)

Deliverables - Overlay tile with `is_goal:true` (on Layer 1). Visual may be placeholder. - Explore Mode detects arrival and triggers a basic win event. **Verification** - Reach Heartroot → win condition fires consistently (even after hazards).

Stage 7 — Interaction rules hardening

Deliverables - Document a one-pager of edge cases: multiple hazards on same cell (not supported in 1.3), sticky + acid precedence (apply mitigation, then damage, then slow registration), repeated damage on re-entry only, etc. **Verification** - Cross-test acid→flesh→acid, sticky→flesh patterns to ensure repeatability.

Stage 8 — QA test maps & checklists

Deliverables - 3 micro-maps (Acid focus, Sticky focus, Digest-gated path) + 1 combined mini-coil ending in Heartroot. - A 10-item checklist the Builder will reuse (e.g., spawn exists, exactly one Heartroot, at least one reachable path, all custom_data keys present on used tiles). **Verification** - All maps pass the checklist; a teammate can reproduce results by following the checklist alone.

Stage 9 — Builder Mode compatibility (pre-work)

Deliverables - Confirm that every 1.3 behavior is **expressible by painting metadata** only (no scripts required in Builder). - Draft the palette spec (names, icons, tooltips, biomass placeholders) so 1.4 can implement the UI directly. **Verification** - Paint a small coil using only TileMap editor; it plays identically to test maps.

Stage 10 — Git & milestone handoff

Deliverables - Branch: `feature/phase-1-3-tiles` → squash merge to `main` after QA. - Tag: `phase-1.3-tilesystem`. - Update CHANGELOG with tiles implemented, metadata schema, interaction order. **Verification** - Clean diff; maps and scenes load on a fresh clone.

6) Risks & Mitigations

- **Stringly-typed metadata:** use limited strings (`tile_type`, `hazard`) and validate presence; add unit sanity checks in Explore Mode during 1.3.
- **Future upgrade combos:** document interaction order now so combo rules plug in later without rewrites.
- **Performance:** metadata lookups are per move; acceptable for MVP. If needed, we'll cache the last tile's data.
- **Builder UX drift:** keep schema stable; any rename must update both TileSet and Explore logic simultaneously.

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7) Acceptance Criteria (Definition of Done)

- All five tile modules function via TileMap metadata only.
- Movement, hazard application, and goal detection follow the documented order.
- QA maps pass; Builder can reproduce maps by painting tiles and setting overlay markers.
- CHANGELOG updated; branch merged; tag created.

Next phase preview (1.4 – Builder Mode): - TilePalettePanel with biomass placeholders reading the same metadata keys. - Validate, Save, Publish buttons (local JSON in 1.5), undo/redo (2.1).